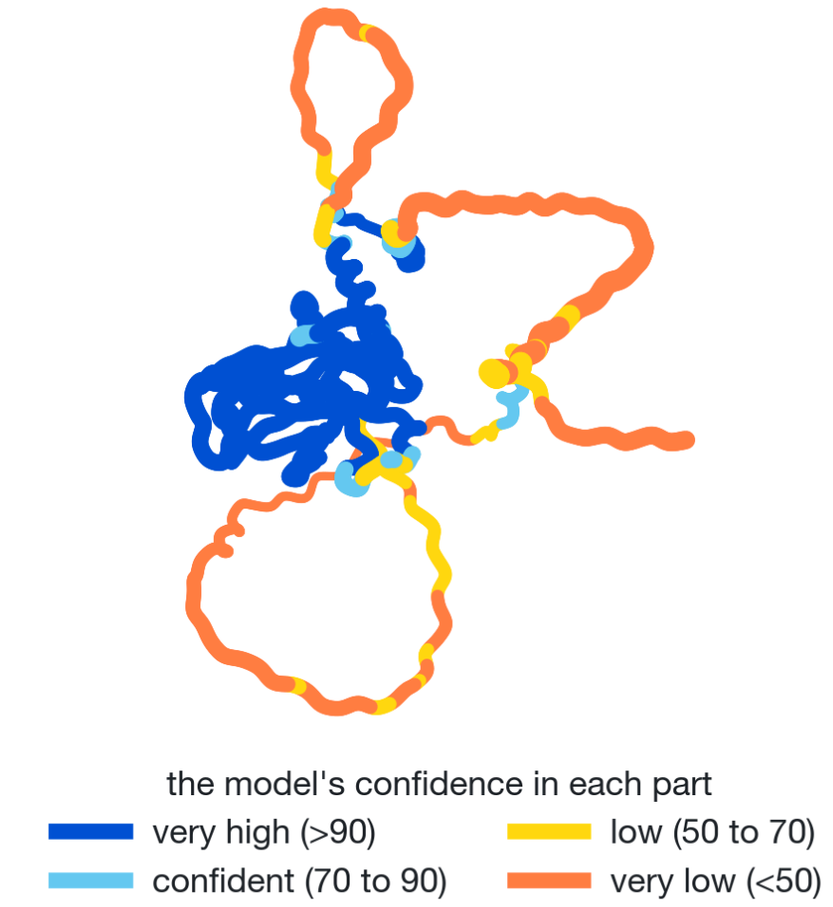
Tutorial 3

CW2, and what this course is for

Friday 30 October 2026

AI for science

- **Nobel Prize in Chemistry 2024**: AlphaFold, predicting protein structure
- **Nobel Prize in Physics 2024**: Hopfield and Hinton, neural networks
- **Since February 2025**, Europe’s weather centre runs a machine-learned forecast in operation



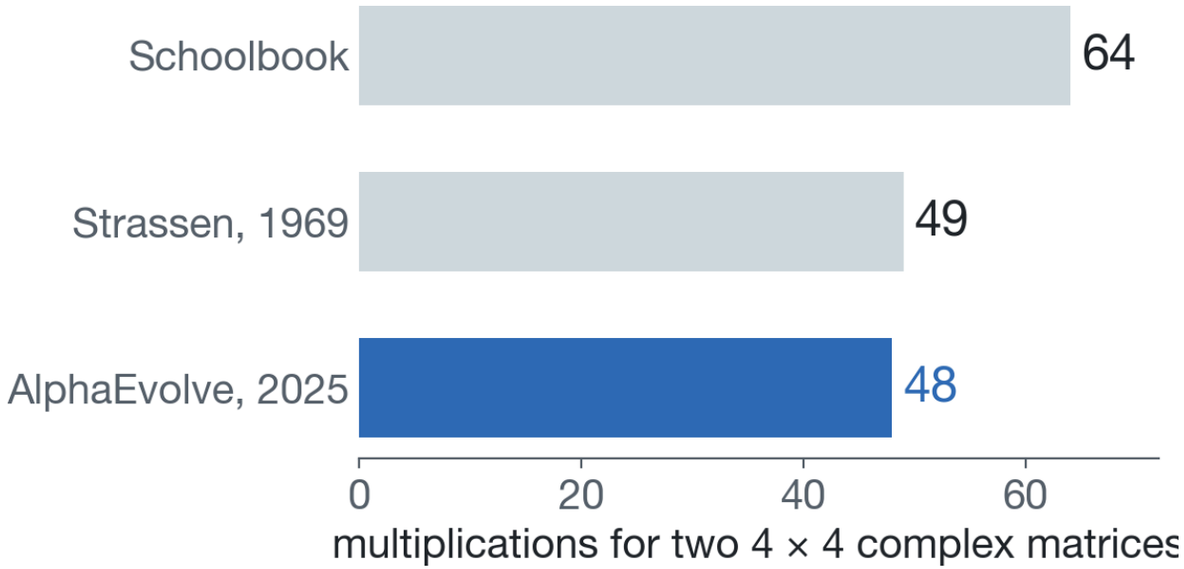
p53, a protein that guards against cancer, as AlphaFold predicts it

Sources: nobelprize.org, 9 Oct 2024; ECMWF, 25 Feb 2025. Structure: AlphaFold DB, AF-P04637, CC BY 4.0

AI finding better algorithms

AlphaEvolve, Google DeepMind, May 2025:

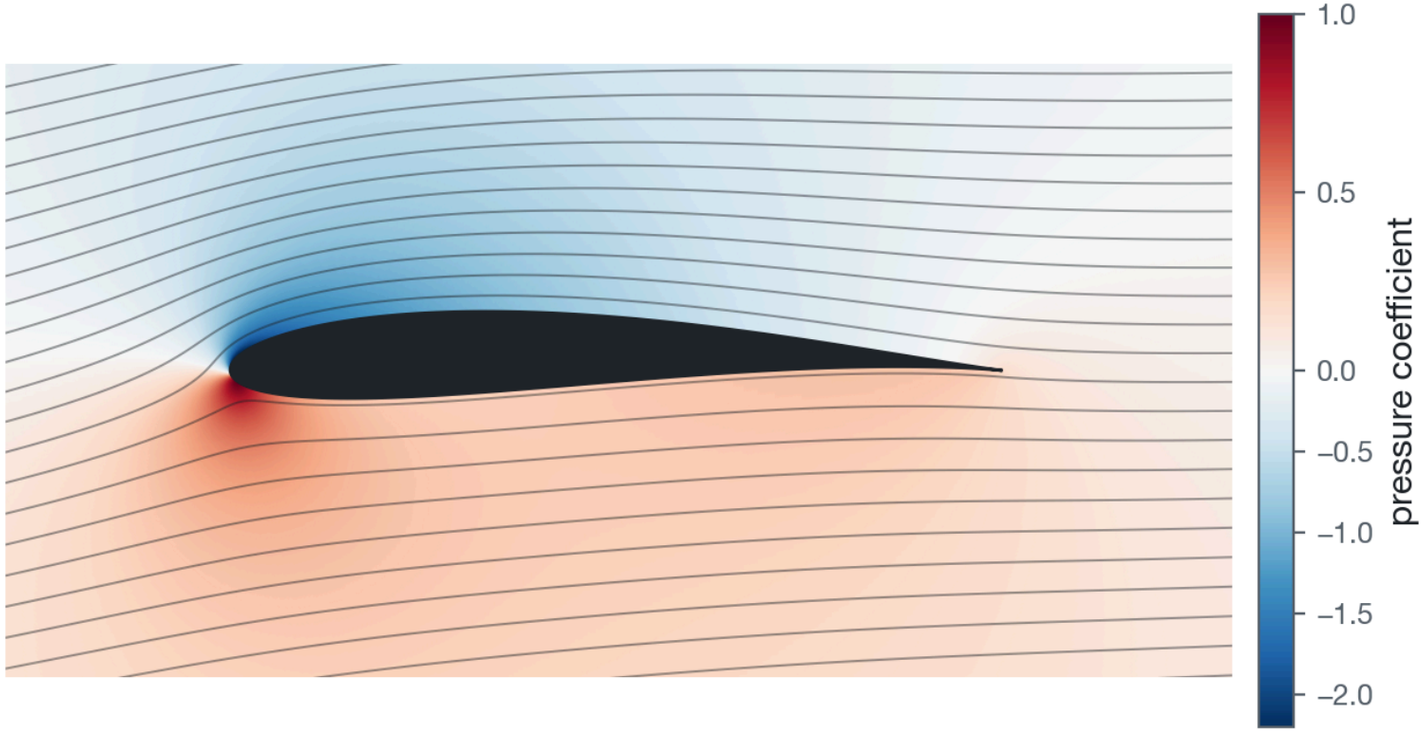
- One multiplication fewer than **Strassen's method from 1969**
- Recovers **0.7%** of Google's worldwide computing
- Sped up the training of **Gemini, the model it runs on**



Source: Google DeepMind blog, 14 May 2025

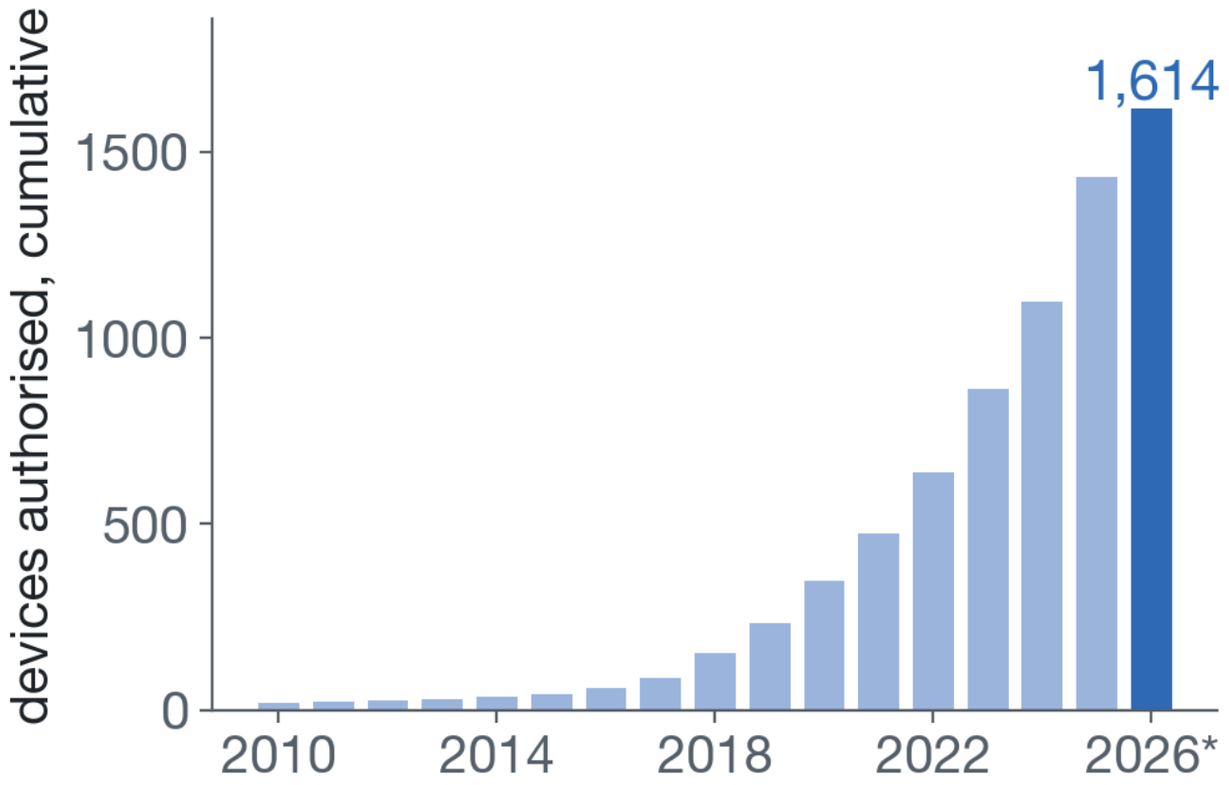
In aerospace

Northrop Grumman: a model trained on CFD simulations predicts thruster plume impingement **“in seconds rather than days”**.



Source: Northrop Grumman, 21 Apr 2026. Figure: potential flow past an aerofoil, drawn for this slide as an illustration, not Northrop’s data

In biomedical engineering



AI-enabled devices authorised by the US FDA. *2026 to the end of June.

Breast screening trial, Sweden, **106,000** women:

- **29%** more cancers found
- **44%** less reading work for radiologists

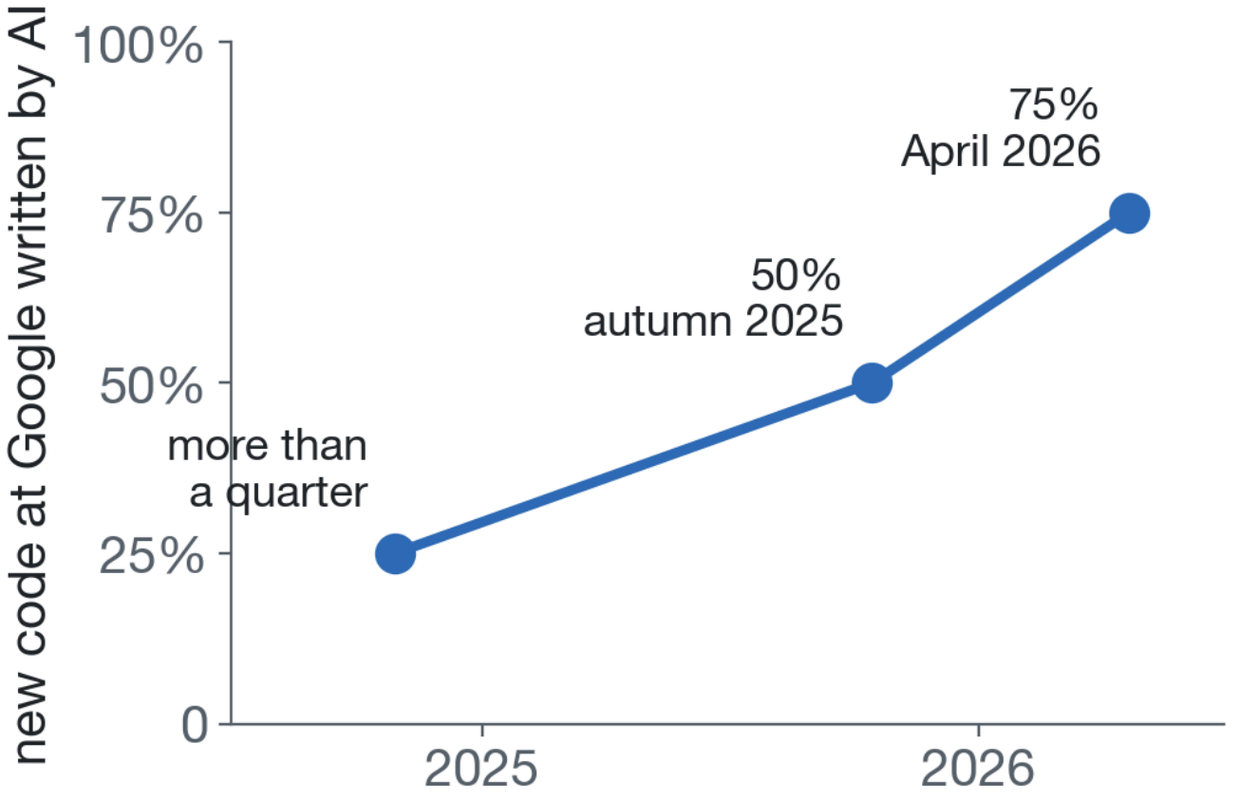
Sources: US FDA list of AI-enabled medical devices, Sep 2026; MASAI trial: Lancet Oncology, Aug 2023; Lancet Digital Health, Feb 2025; The Lancet, Jan 2026

Who writes the code?

75%

of new code at Google is AI-generated, **and approved by engineers**

2025's word of the year: "vibe coding"



Sources: Google, 29 Oct 2024 and 22 Apr 2026; Collins Dictionary, 6 Nov 2025

Tokenmaxxing



Jensen Huang, NVIDIA

> “If that $500,000 engineer did not consume at least $250,000 worth of tokens, I am going to be deeply alarmed.”

- **Meta**: a staff leaderboard of AI use, 60 trillion tokens in 30 days, taken down two days after it leaked
- **Uber**: reported to have spent its 2026 AI coding budget in four months. Its COO, on usage and features shipped: *“That link is not there yet.”*

Sources: All-In podcast via Business Insider, 20 Mar 2026; Fortune, 9 Apr and 26 May 2026. Photo: Office of the President (Taiwan), 2023, CC BY 2.0, Wikimedia Commons

So why am I telling you this?

Most of what you've just seen didn't exist two years ago.

> Every year we ask ourselves: **what is worth teaching**, when the tools change this fast?

What’s left for the engineer

- Framing the **question**
- Choosing **what to measure**
- Knowing when a result **can’t be trusted**
- Weighing **what it costs**

Judgement and taste.

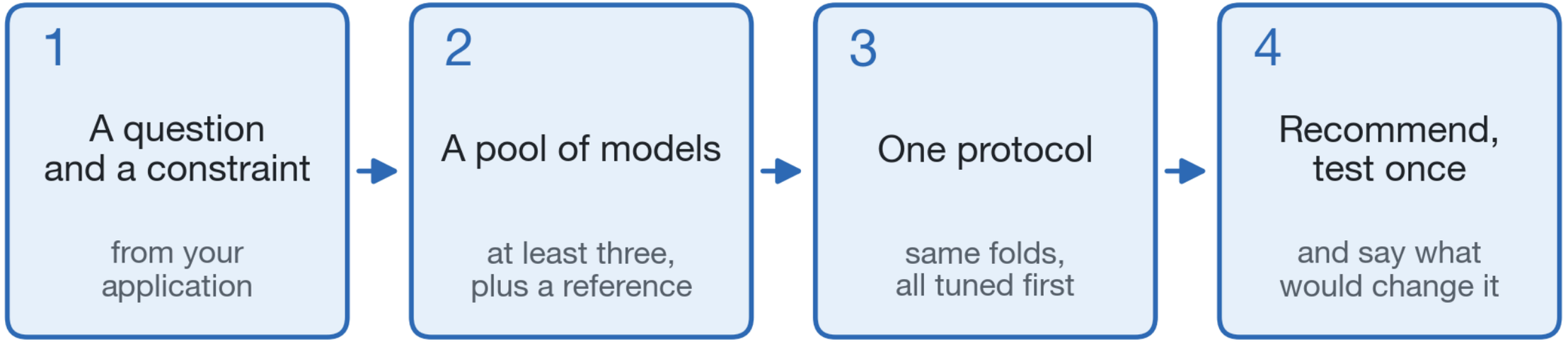
CW2

A recommendation, as an engineering team

CW2 at a glance

- Groups of five, on your dataset
- **40%** of the course: report 20%, presentation 20%
- Submission due **Friday 13 November, 12:00**
- Presentations in the week of **16 November**

The task, in four steps



A constraint comes from the application

- “It must catch at least **80% of failures**”
- “It must score a reading in under **1 ms**”
- “Its prediction must be within **10%** for 90% of cases”

Not a constraint: “as accurate as possible”.

One protocol for every model

- Same split, same folds, same metric
- Every model **tuned first**
- Training score **beside** validation score
- Training and prediction **time**, with the runtime line
- The test set: **once**, at the end

What you submit

- **Report**: up to 8 pages, plus a 1-page appendix, on the template
  - ending with a statement of contribution (not counted)
- **Slides**, as a PDF
- **One group notebook** that reproduces every number in the report
  - a repository link (GitHub) is welcome

The presentation: 10 + 10

- **10 minutes**: question, constraints, comparison, recommendation
- **10 minutes** of questions, which may go to **any member**
- No need to take turns, but make **each member's part** clear

How it’s marked

Report (20%)	Presentation (20%)
Design of experiments and pipelines	Clarity of communication
Accuracy and overfitting	Depth and accuracy of content
Computational efficiency	Justification of the pipeline
Justification against your constraints	Engagement and Q&A

Four criteria each, a quarter each.

Who did what

- One possible split for five: **three model owners**, one **validation** owner, one owner of **testing and cost**
- Background and writing shared
- Individual marks are adjusted only where peer evaluation, the statement of contribution and the Q&A together show a markedly different contribution

To take home

- AI is moving fast, and so is the debate about it
- What's left for you: **judgement**
- CW2 is that judgement, as a team: due **Friday 13 November, 12:00**
- At home: `AI4_T3_compare.ipynb`